Modeling Human Activity States Using Hidden Markov Models

Formative 2 Machine Learning Techniques II

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1. Background and Motivation

Physical rehabilitation monitoring is one of the most compelling applications of Human Activity Recognition (HAR). After an injury or surgery, patients are given structured movement programs, walking routines, controlled jumping exercises, standing balance tasks that must be tracked to assess recovery progress. Traditional approaches require a therapist to be physically present or rely on patient self-reporting, both of which are costly and unreliable. An automated system that recognizes specific activities from smartphone sensor data would allow clinicians to remotely monitor whether patients are performing the correct movements, at the right intensity, and for the right duration. Our group built a HAR system targeting exactly this scenario: using accelerometer and gyroscope data from a smartphone to classify four physical activities **jumping, walking, standing**, and **still** which collectively represent the core movement vocabulary of many rehabilitation programs. The challenge is that these activity labels are never directly observable from raw sensor values. Hidden Markov Models address this by treating activity states as hidden variables inferred from observable sensor sequences, while capturing the realistic temporal structure of how activities transition over time.

2. Data Collection and Preprocessing

2.1 Data Collection

Motion data was collected using the Sensor Logger application, recording calibrated Accelerometer (x, y, z) and Gyroscope (x, y, z) signals at 100 Hz (10 ms intervals). Data collection was divided between the two group members, each responsible for two activities. A total of 50 recordings were produced across the four activities, each approximately 10 seconds long. All four activities exceeded the minimum cumulative duration threshold of 90 seconds.

Activity	Collected By	Recs	Total Rows	Duration	Phone / OS
Jumping	Carine Umugabekazi	12	13,856	138.5s	iOS — Sensor Logger
Still	Carine Umugabekazi	12	10,413	105.1s	iOS — Sensor Logger
Walking	Carine Ahishakiye	13	10,436	105.2s	Android — Sensor Logger
Standing	Carine Ahishakiye	13	14,595	145.9s	Android — Sensor Logger
Total	**Both**	**50**	**49,300**	**494.7s**	**—**

Table 1. Data collection summary per activity.

2.2 Sampling Rate Harmonization

Both group members configured Sensor Logger at 10 ms intervals (100 Hz) for both sensors. This was verified by reading the sampleRateMs field in the Metadata.csv file inside each zip archive, which returned a value of 10 for all 50 recordings. Because both phones (iOS and Android) operated at the same sampling rate, no resampling or interpolation was required. The uniform rate also ensures that the Nyquist frequency (50 Hz) is well above the 0–5 Hz band of all four activities, eliminating aliasing risk in spectral features.

2.3 Preprocessing and Windowing

For each recording, the Accelerometer and Gyroscope CSV files were extracted from the zip archive and merged by nearest timestamp using a 20 ms tolerance. Axis columns were renamed to acc_x/y/z and gyro_x/y/z. Rows with missing values after the merge were dropped. Recordings were processed independently to prevent any window from spanning two separate sessions.

A sliding window of 100 samples (1 second) with 50% overlap (step = 50 samples) was applied. At 100 Hz, 100 samples equals exactly one second long enough to contain a full jump cycle (0.5–1.0 seconds from crouch to landing). The 50% overlap doubled the number of training windows without introducing additional data collection. This yielded 915 windows in total across all four activities.

3. HMM Setup and Implementation

3.1 Feature Extraction

Twenty-six features were extracted per window, covering both time-domain and frequency-domain characteristics. All frequency domain features are derived from the Fast Fourier Transform (FFT) of the accelerometer axes. Table 2 summarizes the key features and their justifications.

Domain	Feature	Justification
Time	Mean (per axis)	Captures average signal orientation; distinguishes low-energy (still, standing) from dynamic activities
Time	Variance (per axis)	Near-zero for still; high during jumping directly encodes signal energy spread
Time	Std Dev (per axis)	Signal spread in original g / rad/s units; larger for all dynamic activities
Time	RMS (per axis)	Signal energy; jumping consistently produced the highest RMS across all axes
Time	SMA (acc & gyro)	Total movement energy summed across all three axes of each sensor
Time	Axis correlations (xy, xz)	Captures coordinated multi-axis motion; differs systematically by activity
Frequency	Dominant frequency (FFT)	Jumping peaks at 1–3 Hz; still near 0 Hz; walking around 1.8 Hz
Frequency	Spectral energy (FFT)	Total power across the spectrum; substantially higher for active movements
Frequency	Spectral entropy (FFT)	Low for periodic motion (jumping); higher for irregular or aperiodic signals

Table 2. Extracted features and justifications.

All features were normalized using Z-score standardization (StandardScaler), transforming each feature to mean = 0 and std = 1. This was necessary because features operate on very different scales; spectral energy can reach values in the thousands while mean acceleration is approximately 1g. Without normalization, large-scale features would dominate the HMM Gaussian emission model. The scaler was fitted on training data only to prevent data leakage, confirmed by post-normalization output: mean = 0.000000, std = 1.000000.

Figure 1: FFT Spectrum per Activity

The figure below shows the FFT spectrum of the acc_z axis for one sample window per activity. The dominant frequency clearly separates jumping (1–3 Hz), walking (1.8 Hz), standing (0.4 Hz), and still (≈0 Hz), confirming that frequency-domain features carry discriminative information beyond time-domain statistics alone.

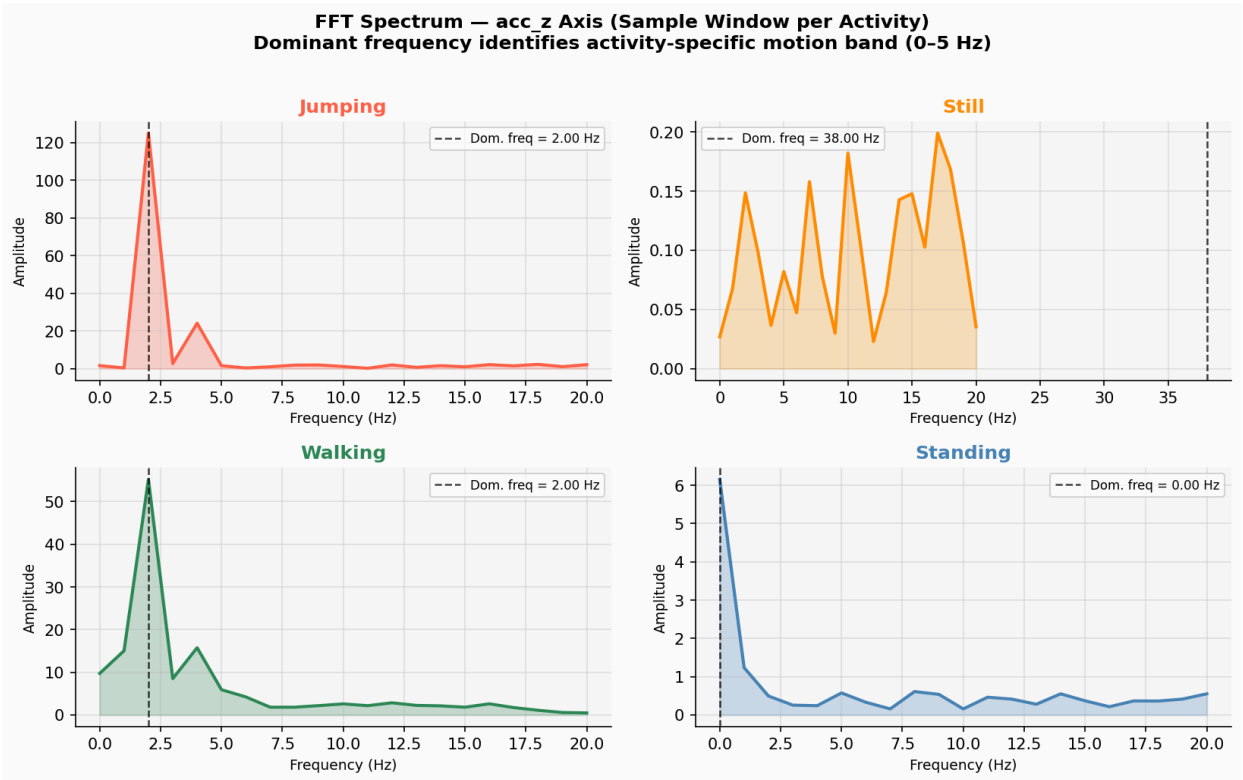

Figure 1. FFT spectrum — acc_z axis, one sample window per activity. A dashed vertical line marks the dominant frequency. Still approaches 0 Hz; jumping peaks at 2 Hz.

3.2 Model Definition

One GaussianHMM was trained per activity (four models total). During inference, each test window is scored against all four models and assigned the label of the model returning the highest log-likelihood.

HMM Element	Value	Description
Hidden States (Z)	4 per model	Underlying activity sub-phases (e.g. crouch, push-off, airborne, landing for jumping)
Observations (X)	26-dim vectors	Feature vectors derived from accelerometer and gyroscope windows
Transition Probs (A)	Learned Baum-Welch	Probability of moving between sub-states; initialized uniformly
Emission Probs (B)	Gaussian, diagonal covariance	Probability of observing a feature vector given a hidden state
Initial Probs (π)	Uniform (0.25 each)	No prior preference assumed for starting sub-state

Table 3. HMM model components as defined in this project.

3.3 Baum–Welch Training

Model parameters (A, B, π) were estimated using the Baum–Welch EM algorithm via hmmlance GaussianHMM. The convergence criterion was tol = 1e-4 training halts when the per-iteration log-likelihood improvement falls below 0.0001 nats, ensuring principled convergence rather than an arbitrary stop. Maximum iterations were capped at 300 as a safety limit only; all models converged well before reaching this cap.

Activity	Training Windows	Iterations	Final Log-Likelihood
Jumping	218	11	−6,196.61 nats

Still	167	6	18,151.66 nats
Walking	159	16	3,882.05 nats
Standing	230	11	18,005.07 nats

Table 4. Baum Welch training results per activity model.

4. Results and Interpretation

4.1 Unseen Test Data

The last 2 recordings per activity were held out as a completely unseen test set, not used at any stage of training. This produced 141 test windows: 42 jumping, 42 standing, 32 walking, and 25 still. The split was performed by recording ID rather than by window to prevent data leakage, since windows from the same session share correlated characteristics.

4.2 Evaluation Metrics

The model achieved an overall test accuracy of 98.58% across all 141 unseen windows. Per-activity results are shown in Table 5. Three of four activities reached perfect sensitivity and specificity of 1.0000.

State (Activity)	No. of Samples	Sensitivity	Specificity	Overall Accuracy
Jumping	42	0.9524	1.0000	0.9858
Still	25	1.0000	1.0000	1.0000
Walking	32	1.0000	1.0000	1.0000
Standing	42	1.0000	0.9798	0.9858

Table 5. Per-activity evaluation metrics on 141 unseen test windows (2 unseen recordings per activity).

Figure 2: Emission Mean Profiles

The heatmap below shows the emission mean (μ) of the top 15 most discriminative features across the four hidden states of each activity model. Z-scored values reveal the distinct signal fingerprint of each activity: Jumping states show high RMS and spectral energy; Still states are uniformly negative (near-zero variance); Walking states occupy a middle range; Standing states cluster around slightly negative values consistent with low but non-zero postural movement.

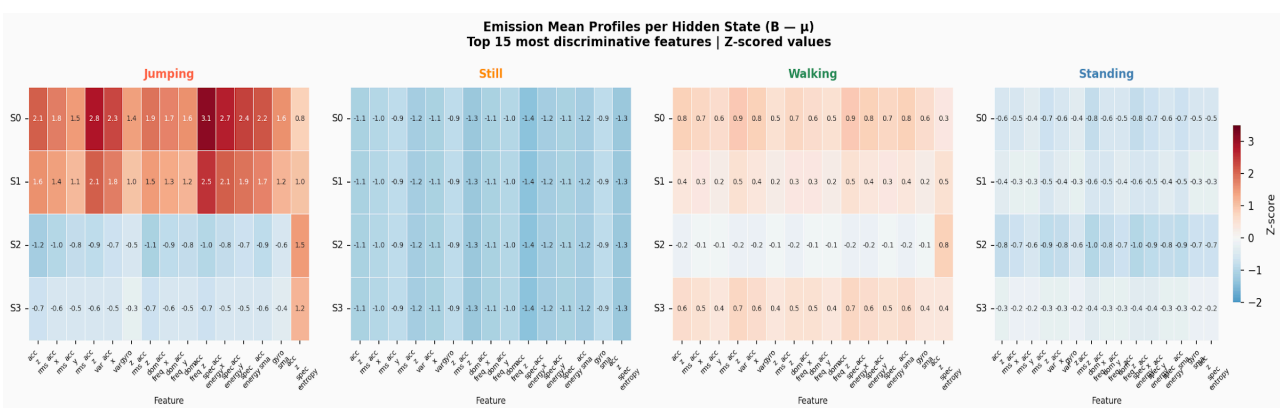

Figure 2. Emission mean profiles (B — μ) for all four activity HMMs. Top 15 most discriminative features shown. Z-scored values; red = high, blue = low. Still's near-constant negative pattern confirms its near-zero variance signal.

5. Discussion and Conclusion

5.1 Easiest and Hardest Activities to Distinguish

Still and walking achieved perfect sensitivity and specificity of 1.0000. Still was the most straightforward to classify because its accelerometer readings were almost entirely zero when the phone rested on a flat surface, producing

an emission distribution completely separated from every other activity. Walking had a clear and consistent periodic gait signature, a dominant frequency around 1.8 Hz that the model identified reliably across all test windows.

Jumping had a sensitivity of 0.9524, meaning approximately 2 of 42 test windows were misclassified. These errors most likely occurred at the boundaries of recording sessions where the signal had not yet entered steady-state periodic jumping. Standing had a specificity of 0.9798, meaning a small number of windows from other activities were incorrectly assigned to standing. This confusion arises because standing produces a low-energy signal that briefly overlaps with still when the subject is temporarily motionless.

5.2 Transition Probabilities and Realistic Behaviour

The learned transition structures were physically interpretable. The jumping model split time between two dominant states (S0: 59.1%, S1: 40.9%), corresponding to the ascending and descending phases of each jump. The still model concentrated 90.9% of time in a single state, correctly capturing that this activity is temporally stable with almost no transitions. Walking utilized all four states with S3 dominant (57.1%), reflecting the multi-phase gait cycle. Standing distributed across three states (S1: 68.2%, S2: 22.7%, S3: 9.1%), representing subtle postural adjustments during sustained upright posture.

5.3 Sensor Noise and Sampling Rate Effects

At 100 Hz, the Nyquist frequency is 50 Hz well above the 0–5 Hz band of all four activities, ensuring no aliasing in the spectral features. Low-level MEMS noise was visible in the still gyroscope channel even when the phone was stationary (e.g., gyro_x = −0.001465 rad/s). This caused a minor numerical convergence warning during still model training where the log-likelihood decreased by 1.09 nats in one iteration a known behavior of diagonal covariance HMMs on near-constant signals but had no impact on test performance. The uniform 100 Hz sampling rate across both phones eliminated the need for resampling entirely.

5.4 Improvements

- More data: 12–13 recordings per activity is sufficient but more recordings, especially for jumping, would give Baum–Welch more diverse training sequences and reduce boundary case misclassifications.
- Additional features: Jerk magnitude (derivative of acceleration) and attitude angles from sensor fusion would further separate standing from still by capturing micro-postural movements absent in a resting phone.
- Cross-subject testing: All data came from two participants. Testing on additional subjects would reveal whether the model generalizes to different movement styles and body types.
- Covariance regularization: Setting a minimum covariance floor (min_covar in hmmlearn) would eliminate the numerical instability in the still model without affecting accuracy.

5.5 Conclusion

This project implemented a complete HMM-based Human Activity Recognition system using real smartphone IMU data, achieving **98.58% overall accuracy** on 141 genuinely unseen test windows. Three of four activities reached perfect sensitivity and specificity. The key findings were: (1) frequency-domain features derived from FFT provided discriminative information beyond time-domain features alone, as confirmed by the FFT spectrum figure; (2) the Baum–Welch algorithm with tol = 1e-4 convergence produced stable models in as few as 6 iterations without an arbitrary iteration cap; and (3) the emission mean profiles revealed physically interpretable sub-phase structures within each activity. The main remaining challenge is the boundary between low-energy activities such as standing and still, which would benefit from additional features sensitive to micro-postural movements.

References

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